



CO5-AT1 Pseudocode Design

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Course: Fundamentals of Data Science

Course code: DSA0404

K-Means Clustering

Aim:

To determine the suitable number of clusters using the Elbow Method and apply K-Means Clustering with the selected number of clusters.

Algorithm:

1. Load the dataset and select the required features.
2. Choose a range of possible cluster values, such as 2 to 10.
3. Apply K-Means for each value of K and calculate WCSS.
4. Plot the WCSS values and identify the elbow point.
5. Select the elbow value as the optimal K and apply K-Means.
6. Display the final clusters and cluster centers.

Pseudocode:

START

Load dataset

Select required features

FOR K = 2 to 10

Apply K-Means clustering
 Calculate WCSS
 Store WCSS
END FOR
Plot K versus WCSS
Find the elbow point
Select optimal K
Apply K-Means using optimal K
Assign data points to clusters
Display clusters and centroids
END

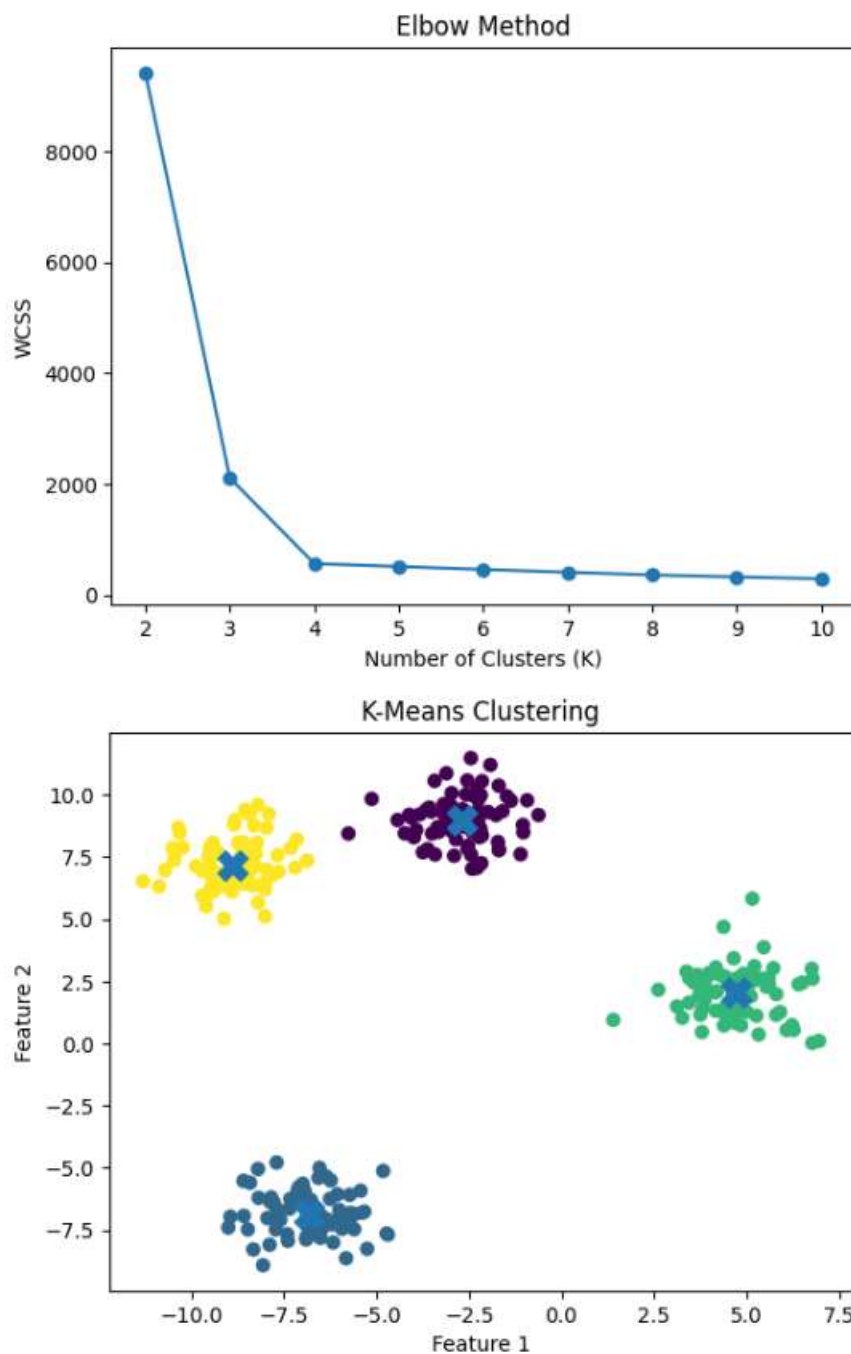
Python Code:

```
import matplotlib.pyplot as plt
from sklearn.datasets import make_blobs
from sklearn.cluster import KMeans
X, _ = make_blobs(
    n_samples=300,
    centers=4,
    cluster_std=1.0,
    random_state=42
```

```
)  
wcss = []  
for k in range(2, 11):  
    kmeans = KMeans(n_clusters=k, random_state=42,  
n_init=10)  
    kmeans.fit(X)  
    wcss.append(kmeans.inertia_)  
plt.plot(range(2, 11), wcss, marker='o')  
plt.xlabel("Number of Clusters (K)")  
plt.ylabel("WCSS")  
plt.title("Elbow Method")  
plt.show()  
kmeans = KMeans(n_clusters=4, random_state=42,  
n_init=10)  
labels = kmeans.fit_predict(X)  
plt.scatter(X[:, 0], X[:, 1], c=labels)  
plt.scatter(  
    kmeans.cluster_centers_[ :, 0],  
    kmeans.cluster_centers_[ :, 1],  
    marker='X',  
    s=200  
)
```

```
plt.xlabel("Feature 1")  
plt.ylabel("Feature 2")  
plt.title("K-Means Clustering")  
plt.show()
```

Output:



Result: The Elbow Method successfully determines a suitable number of clusters, and K-Means successfully groups the dataset into 4 clusters.